

AYUSH UPADHYAY

Chennai, India • +91-9305183418 • ayushdevxai@gmail.com
linkedin.com/in/ayushdevai • github.com/batmandevx • ayushxupadhyay.netlify.app • leetcode.com/u/N5uq6wDQyn

SUMMARY

Computer Science (AI & ML) undergraduate, class of 2027, with seven software engineering and applied research roles across Microsoft, Calyptus, Nokia, IIT Ropar, and FlowbitAI Germany. Ships production systems end to end: LLM and RAG pipelines, agentic automation, distributed streaming and inference backends, multithreaded C++ systems tooling, and full-stack web applications. Strong foundation in data structures, algorithms, object-oriented design, and scalable system design. 3 IEEE publications, 3 patents, 17 hackathon wins including 2nd place at Tata InnoVent 2026 across 2,822 teams. Seeking 2027 Software Engineering / AI Engineering internship and new-grad developer roles.

EDUCATION

B.Tech, Computer Science and Engineering (AI & Machine Learning)

2023 – 2027

Vellore Institute of Technology, Chennai, India

CGPA 8.5 / 10

- Coursework: Data Structures and Algorithms, Operating Systems, Computer Networks, Database Management Systems, Object-Oriented Programming, Machine Learning, Computer Architecture, Software Engineering.
- Leadership: Founder, HackFinity (1,550+ participants, 65+ countries) • Chairperson, Institution of Engineers India (IEI) VIT Chennai • Technical Lead, ACM Student Chapter.

EXPERIENCE

Applied Scientist Intern | Microsoft, Hyderabad, India

May 2026 – Jul 2026

- Built and shipped a cohort-based recommendation engine for Microsoft 365 Copilot upskilling inside Viva Learning, clustering users on behavioral and skill signals to serve adaptive course recommendations from a user's first day of activity.
- Designed the cohort-vector schema and a five-generation synthetic data pipeline with automated quality validation, registering datasets and training runs on Azure ML.
- Benchmarked nine clustering approaches in a segmentation prototype, then integrated the selected model into AILA, the product surface consuming the engine.

Junior AI Engineer | Calyptus (AI recruitment startup), Remote

May 2026

Promoted from AI Intern, Apr 2026 – May 2026

- Built the Warm Lead Network Monitor in 72 hours during the first week: a Python service that extracted and deduplicated 400+ warm contacts from the founders' all-time email archives and turned a cold outbound motion into a warm one.
- Automated continuous tracking of every contact's company careers page for new job openings, with change detection and a weekly digest reporting company, role count, and contact details.
- The digest now drives weekly outbound for the CEO and co-founder and converted into 3 booked demo calls; featured as company "Goat of the Week" for the delivery.

Full Stack AI Developer | Tenchi Consultation, Dubai, UAE

Jan 2026 – Mar 2026

- Architected SmartOrder Engine, an AI order-management layer over SAP S/4HANA serving FMCG distributors across the UAE, Qatar, and Saudi Arabia.
- Delivered automated invoice OCR, real-time inventory tracking, and Gemini 2.0 Flash demand forecasting behind a REST API consumed by client ERP workflows.

AI Full Stack Developer | FlowbitAI, Germany

Jan 2026 – Mar 2026

- Redesigned the document OCR pipeline around a CNN-Transformer hybrid, raising extraction accuracy 35% while cutting inference cost 40%.
- Rebuilt the backend as an async FastAPI service with Redis queuing and AWS Lambda workers, holding sub-2-second latency at 50,000+ pages per month.

AI Developer Intern | Nokia

May 2025 – Aug 2025

- Deployed XGBoost and LSTM predictive-maintenance models across 12 5G/6G network clusters, reducing unplanned downtime 30%.
- Built event-driven Apache Kafka streaming pipelines ingesting 1M+ telemetry records daily into the feature store powering those models.

AI/ML Research Intern | Annam AI, IIT Ropar

Apr 2025 – Nov 2025

- Trained computer vision models in TensorFlow and PyTorch to 92% accuracy on limited-sample agricultural datasets.
- Accelerated training convergence 40% through transfer learning and augmentation strategies on pretrained backbones.

AI Software Engineer Intern | Real IT Solutions

May 2025 – Jul 2025

- Shipped full-stack AI applications in React, Node.js, and FastAPI sustaining 99.9% uptime.
- Automated CI/CD with GitHub Actions and Git-based code review workflows, cutting deployment cycle time 85%.

PROJECTS

NeuroDrive — Agentic Self-Healing EV Ecosystem | Python · Agentic AI · AWS · IoT

- 2nd place at Tata InnoVent 2026 among 2,822 teams and 10,247 innovators. Autonomous EV diagnostics with an agentic self-healing framework that cut simulated fleet downtime 65%, plus real-time fleet monitoring on AWS.

Packet Analyzer — Deep Packet Inspection Engine | C++ · Python · Multithreading · TLS/SNI · BPF

- Multithreaded DPI engine with SNI/TLS extraction, BPF-filtered capture, per-flow state tracking, and a load-balanced worker pipeline for live traffic classification.

CrispRoots — AI AgroTech Platform | Machine Learning · Computer Vision · IoT · React

- Won Rs 2L at the IIT Ropar Annam AI Hackathon plus Rs 1L seed funding from Balkalman Funds. Precision agriculture platform with crop disease detection, soil analysis, and yield prediction.

GyanGrow — Gamified Financial Literacy Platform | React · Node.js · TensorFlow · MongoDB

- 50,000+ users and 100+ quests with AI-personalized learning roadmaps, tracking Rs 2Cr+ in virtual savings at a 95% quest completion rate.

Echo-Med-AI — Mobile Medical Diagnostics | TensorFlow · OpenCV · CNN · TFLite

- CNN diagnostic tool reaching 94% accuracy across six disease categories, optimized for mobile inference via TFLite quantization with a 75% reduction in model size.

AuraTwin — AI Infrastructure Safety | AWS Bedrock · Gaussian Splatting · 3D Reconstruction

- Entry in the AWS 10,000 Aldeas Competition. Phone-based structural inspection using Gaussian Splatting for 3D reconstruction and Amazon Bedrock for real-time risk analysis.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, JavaScript, TypeScript, Go, Rust, Kotlin, SQL, Bash, HTML/CSS

CS Fundamentals: Data Structures, Algorithms, Dynamic Programming, Graph Algorithms, Recursion and Backtracking, Time and Space Complexity Analysis, Object-Oriented Design, Design Patterns, Operating Systems, Computer Networks, DBMS, Compiler Basics

System Design: High-Level and Low-Level Design, Distributed Systems, Scalability, Sharding, Replication, Consistent Hashing, CAP Theorem, Load Balancing, Caching Strategies, API Gateways, Message Queues, Event-Driven Architecture, Rate Limiting, Fault Tolerance, Concurrency and Multithreading, Low-Latency Optimization

AI / ML: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, OpenCV, Transformers (BERT, GPT), Hugging Face, CNNs, LSTMs, Feature Engineering, Hyperparameter Tuning, Transfer Learning, Model Evaluation, Quantization, Model Deployment, MLOps

LLM and GenAI: LangChain, LangGraph, LlamaIndex, RAG, Vector Databases, Embeddings, Semantic Search, Fine-Tuning, LoRA/QLoRA, Prompt Engineering, LLM Orchestration, AI Agents, Agentic AI, Web Scraping and Automation (Playwright, BeautifulSoup), Voice AI (ASR/TTS), NLP, Computer Vision

Backend and Web: React, Next.js, Node.js, Express, FastAPI, Django, Flask, REST API Design, GraphQL, Microservices, WebSockets, JWT and OAuth Authentication, Rate Limiting, Async Programming, Server-Side Rendering

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Pinecone, Query Optimization, Indexing, Schema Design, Data Modeling

Cloud: AWS (EC2, S3, Lambda, SageMaker, Bedrock, IAM, CloudWatch, API Gateway), Azure (Azure ML, Blob Storage), Serverless Architecture, Cloud Cost Optimization

DevOps and Infrastructure: Docker, Kubernetes, CI/CD Pipelines, GitHub Actions, Jenkins, Nginx, Terraform, Infrastructure as Code, Container Orchestration, Apache Kafka, Monitoring and Observability, Prometheus, Grafana, Log Aggregation, Blue-Green Deployment, Linux, Shell Scripting, Git and GitHub, Version Control

Testing and Quality: Unit Testing, Integration Testing, pytest, Jest, Debugging, Profiling, Code Review, Test-Driven Development

Data and Tools: Microsoft Excel (advanced formulas, pivot tables, VLOOKUP, data modeling, dashboards), Pandas, NumPy, Matplotlib, Jupyter, Power BI, Postman, SAP S/4HANA, Figma, Jira, Confluence

Practices: Agile and Scrum, Sprint Planning, Technical Documentation, Cross-Functional Collaboration, Stakeholder Communication, Mentoring, Problem Solving, Ownership

PUBLICATIONS AND PATENTS

- Patent (Granted, India) — No. 202541101828 A — “Regenerative Power System for Vehicles,” an AI-orchestrated quantum-tunnelling braking system.
- Patent (Filing in progress) — Advanced Selective Active Noise Cancellation System: adaptive, selective suppression of targeted noise bands.
- Patent (Filing in progress) — BB-KDF, a Behavioral Biometric Key Derivation Function deriving ephemeral AES-256 session keys from live keystroke, pointer, touch, and gaze signals via LSTM, Reed-Solomon, and HKDF.
- IEEE CICT 2025 — “Efficient Pruning in Subset Sum Backtracking” (Paper ID CICT2025-540).
- IEEE — “AI-Driven Resource Optimization for 5G/6G Heterogeneous Networks”: spectrum allocation and predictive load balancing.
- IEEE — “Solid-State Battery Degradation Prediction via Hybrid ML”: LSTM-attention for cycle-life estimation.

ACHIEVEMENTS AND LEADERSHIP

- 17x Hackathon Winner — Tata InnoVent 2026 (2nd place, 2,822 teams), Samsung Solve for Tomorrow, Annam AI IIT Ropar, HackHazards, AWS 10K Aldeas, IntelliTrace (Indian Bank), Garage Day, and 10 more.
- Innovation Ambassador, Ministry of Education, Government of India.
- Organizer, Nexora Hacks 2026 — \$7,500 prize pool, 695 participants, 32 judges from Apple, Meta, Netflix, AWS, and Walmart.
- Certifications: Generative AI (IBM) • AI Fundamentals (Microsoft) • Full Stack Developer (Coursera) • AI Developer (AMD) • AWS Cloud Practitioner • GitHub Professional • LangChain and LangGraph • Backend Development.